# File Handling Operations in Python

A comprehensive guide to working with files in Python


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PRN NUMBER: [202501040118

# What is File Handling?

File handling allows programs to **read from** and **write to** external files

- Store data permanently
- Process large datasets
- Save program state
- Import/export information

# Types of Files

## Text Files

Human-readable format

- .txt, .csv, .json
- Plain text data
- Can be opened in any text editor

## Binary Files

Machine-readable format

- .exe, .png, .mp3
- Stored as bytes
- Require specific programs to read

# File Modes in Python

Each mode determines how you interact with the file

## r - Read

Open for reading (default)

## w - Write

Open for writing (overwrites)

## a - Append

Open for adding to end

## x - Create

New file or error if exists

## b - Binary

Binary mode flag

## +

Read and write combined

# Basic File Operations

- **open()**
  Creates file object
- **read()**
  Reads file content
- **write()**
  Writes to file
- **close()**
  Frees resources

# Reading Methods

**01**

## read()

Returns entire file as string

**02**

## readline()

Returns one line at a time

**03**

## readlines()

Returns list of all lines

# Writing to Files

## write()

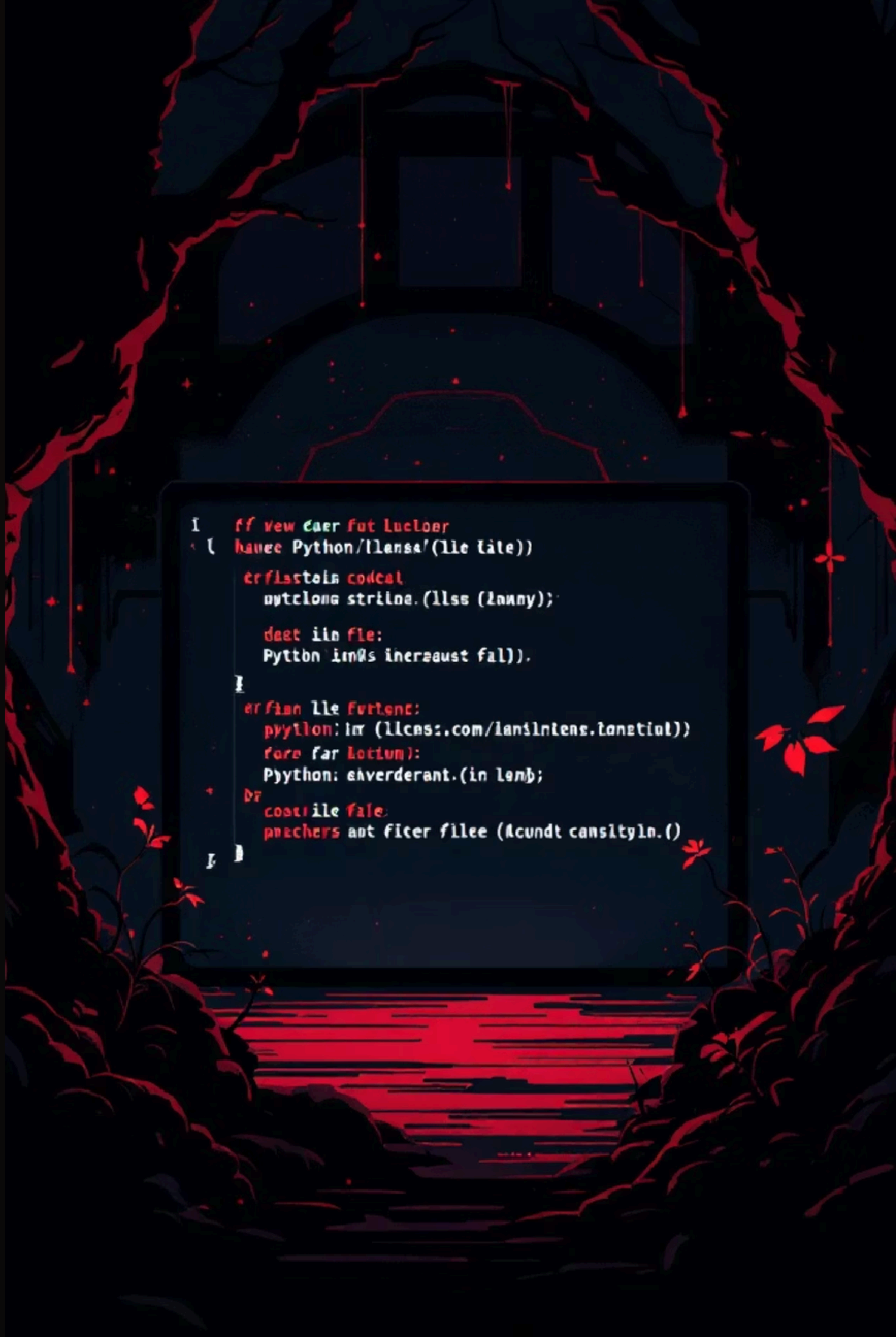
Writes a string to the file

Returns number of characters written

## writelines()

Writes a list of strings

No newlines added automatically



# Example Program

Write a Python program to create a file, write data, and read it back

```
# Create and write to file
file = open("example.txt", "w")
file.write("Hello, World!\n")
file.write("This is Python file handling.\n")
file.close()

# Read from file
file = open("example.txt", "r")
content = file.read()
print(content)
file.close()
```

**Output:** Hello, World! This is Python file handling.

# Advantages of File Handling

### Data Persistence

Data survives after program ends

### Data Sharing

Easy data exchange between programs

### Configuration

Store settings and preferences

### Large Data

Process datasets too big for memory

# Key Takeaways

**1** **Master the modes**

Choose the right mode for each operation

**2** **Always close files**

Use context managers for automatic cleanup

**3** **Handle exceptions**

Plan for file not found errors

Happy Coding!